

QUESTION-21

HEAP SORT

```
#include <stdio.h>
void heapify(int a[], int n, int i)
{
    int largest = i; int l = 2 * i + 1; int r = 2 * i + 2; int temp;
    if (l < n && a[l] > a[largest])
        largest = l;
    if (r < n && a[r] > a[largest])
        largest = r;
    if (largest != i)
    {
        temp = a[i];
        a[i] = a[largest];
        a[largest] = temp;
        heapify(a, n, largest);
    }
}
void heapSort(int a[], int n)
{
    int i, temp;
    for (i = n / 2 - 1; i >= 0; i--)
        heapify(a, n, i);
    for (i = n - 1; i > 0; i--)
    {
        temp = a[0]; a[0] = a[i]; a[i] = temp; heapify(a, i, 0);
    }
}
```

Sorted Array: 12 23 29 34 56 78

=== Code Execution Successful ===

```
int main()
{
    int a[] = {23, 56, 12, 34, 78, 29}; int n = 6, i;
    heapSort(a, n);
    printf("Sorted Array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
```